



Regular article

Experimentally validated defect depth estimation using artificial neural network in pulsed thermography



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ARTICLE INFO

Keywords:

Pulsed thermography
Non destructive evaluation
Finite elements method
Neural network
Defect depth detection

ABSTRACT

Infrared thermography (IRT) proved to be a valuable technique in non-destructive testing due to its contactless nature, real-time application, and wide area coverage. However, the major hurdle to widespread application of IRT is quantification; more specifically quantifying the depth of defects from the acquired thermograms in low conductive host materials. In this paper, an artificial neural network (NN) is employed to detect defects depth in composite samples, coupled with a Pulsed Thermography PT setup, to complement prior work when NN algorithms were coupled to a line-scan thermography. Carbon Fiber Reinforced Polymer (CFRP) coupons with embedded and flat bottom holes defects were designed via 3D printing; to precisely control the defects morphology and depth. Firstly; the current study presents a proof of concept using a Multiphysics FEM simulation model of the inspection process, to generate the training sets of data, so that the developed NN is assessed in a deterministic (noise free) environment. Then, the proposed NN was further tested experimentally to validate its accuracy and performance. The accuracy of the developed NN for the synthetic data was more than 97% and for the experimental data was around 90%.

1. Introduction

Non-destructive testing (NDT) has emerged as a non-invasive technology for examining the integrity and robustness of high-performance composite structures such as Carbon Fiber Reinforced Polymers (CFRP), especially when applied in wind turbine blades and multi-layered aeronautical structures that require recurring inspection routines to avoid any catastrophic failure [1]. More specifically, defects in the form of voids, disbonds, delamination and cracks must be revealed to ensure their state of health [2–4]. Several NDT methods are available to reveal such defects that include, but not limited to, ultrasonic testing [5], Infrared Thermography [6], X-ray [7], Laser [8] and Super Conductive Quantum Interference Devices (SQUID) [9], where ultrasonic testing is one of the oldest and well-established methods. Other alternatives to ultrasonic testing have emerged due to ultrasound limitations; requires intimate contact with the surface, and has slow scanning speeds; thus covering small area per unit time. Infrared thermography, on the other hand, is superior in this aspect as it is a contactless and speedy process, allowing for a wide area coverage inspection [10]. At the same time, thermography suffers from other limitations, mainly its limited depth of penetration, which restricts its usage to thin samples that are usually less than 10 mm in thickness [11]. This drawback can be explained by the attenuation of the defect thermal signature as the

depth increases. It is clearly obvious that there is a need for a set of robust image enhancement techniques and computational methods to reliably process the acquired thermal response; and help extract the defects' characteristics in terms of size and depth, respectively.

Several techniques were implemented to preprocess the acquired thermograms and improve defects detection; such as Principal Component Analysis (PCA) [12–14], Self-referencing [15], Thermographic Signal Reconstruction (TSR) [16,17], a combination of automatic seeded region growing and TSR [18] and many more. All these techniques when employed, enhance the visibility of the deeper defects or highlight defects that were not visible in raw frames. Although this can be quite helpful for qualitative assessment of composite structures, however, this may not be adequate for conducting a quantitative study in which the goal is to estimate the defect size and depth.

Quantitative analysis becomes particularly more challenging in materials that have anisotropic thermal properties such as composite structures. One dimensional heat conduction equation has been used in the literature in an attempt to model the thermal response of samples under study [19]. An extension to two-dimensional models was also conducted using the finite difference method examined by [20]. Three-dimensional modeling of composite structures is the most accurate and most reliable. It is being investigated nowadays using FEM computer simulation environment such as COMSOL® [21,22,6].

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Received 20 January 2019; Received in revised form 19 February 2019; Accepted 12 March 2019

Available online 16 March 2019

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Several methods have been proposed to estimate the defect depth in pulsed thermography using a specific characteristic time, which varies per the method used. Peak temperature contrast approach makes use of the thermal contrast curve and uses the peak temperature time as a characteristic time for estimating the depth using the one-dimensional heat conduction solution [23]. It was found that this characteristic time is proportional to the square of the defect depth and the proportionality constant is dependent on the lateral size of the defect. JG Sun [24] reported that this method is unable to estimate the defect depth. Peak temperature slope contrast method on the other hand uses the peak time of the first derivative of the thermal contrast curve for depth prediction [24]. This approach, however is unable to estimate defects' depths that are comparable to the sample thickness. The least square fitting method is another technique for depth prediction that is less sensitive to noise [25]. Despite of its accurate depth prediction for low thermal conductivity material, this method is not suitable for high thermal conductivity materials such as metal specimens. Pulsed phase thermography algorithm analyzes the thermal response in the frequency domain for depth detection. This method, however, is less accurate for shallow defects [26].

In this research, we first modeled a 3D based CFRP specimen in COMSOL®. The model simulates heat source of a pulsed thermography setup over the inspected test specimen. The thermal response was simulated using the numerical model, which in turn was used to train the artificial neural network to estimate the defects' depths. As the simulated data is noise-free, we further validated the robustness by testing the proposed neural network approach experimentally. This proposed work is meant to further highlight the applicability and associated accuracies with the NN-based post-processing of pulsed thermograms. Our prior work in implementing NN codes to quantify line-scan thermograms presented similar principles, but with different heating/stimulation physics (rising and falling curves). The combination of our studies of NN application to line-scan (details in reference [28]) and pulse-thermography (current study), should provide insights on the process of extracting training sets among other important, yet context-sensitive, deployment practices (such as standardization).

2. Pulsed thermography

In pulsed thermography, a short and high-power thermal pulse is applied to the specimen via heat radiation. Due to the heat conduction, heat absorbed by the surface will propagate through the material. The absorbed heat creates a thermal front, which while its propagation can be altered by any internal defects (variation in thermal conductivity) in the material. As the time elapses, the surface temperature of the specimen will decrease uniformly for a defect-free zone. On the other hand, a surface spot having an internal defect underneath it will have a higher temperature, with a temperature decay that is dissimilar to the sound area. This is due to the resistance (lower thermal diffusivity) to the heat flow caused by an internal defect. This difference between the temperatures can be observed using an infrared camera. Fig. 1 shows the basic working principle of pulsed thermography.

3. FEM simulations

Finite Element Method simulations proved to be a fast and an economical way to examine the thermal response of pulsed thermography [22,27]. It is specifically of interest to utilize 3D based simulation as the study is considering CFRP composite structures, which have anisotropic thermal properties and low conductivity. The heat transfer module in COMSOL® was incorporated in order to solve the 3D transient energy equation to obtain the temperature curves of the CFRP surface, as represented in Eq. (1);

$$\rho C_p \left(\frac{\partial T}{\partial t} \right) - \nabla \cdot (k \nabla T) = 0 \quad (1)$$

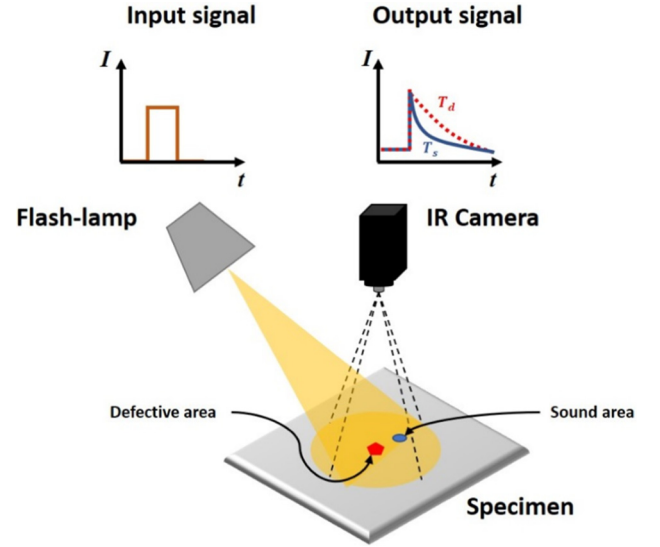


Fig. 1. Working principle of Pulsed Thermography.

where $\rho \left(\frac{\text{kg}}{\text{m}^3} \right)$ is the density, $C_p \left(\frac{\text{J}}{\text{kg} \cdot \text{m}} \right)$ is the heat capacity at constant pressure, $T(\text{K})$ is the absolute temperature, $k \left(\frac{\text{W}}{\text{m} \cdot \text{K}} \right)$ is the material thermal conductivity and $t(\text{s})$ refers to time. The ambient temperature of the environment is set to 293.15 K as an initial condition for this simulation. The boundary condition on the other hand is set by the convection and radiation of heat from the surface of the CFRP sample, which is described in Eq. (2) below:

$$n(k \nabla T) = q_o + h_{\text{conv}}(T_{\text{amb}} - T) + \sigma \epsilon (T_{\text{amb}}^4 - T^4) \quad (2)$$

where $q_o \left(\frac{\text{W}}{\text{m}^2} \right)$ is the heat source, $h_{\text{conv}} \left(\frac{\text{W}}{\text{m}^2 \cdot \text{K}} \right)$ is the constant convective heat transfer coefficient, $\sigma \left(\frac{\text{W}}{\text{m}^2 \cdot \text{K}^4} \right)$ is the Stefan-Boltzmann constant, ϵ is the emissivity. The last two terms account for the heat flux due to convection and radiation respectively.

CFRP structures are comprised of bi-directional woven fibers stacked on top of each other. Modeling their heat-flow requires defining the heat flow in an anisotropic fashion and the use of curvilinear physics (and thus curvilinear coordinates as well), which is computationally expensive in terms of processing time and power requirements; due to the fine meshing. To overcome this problem, the study employed the equivalent thermal conductivity (as explained thoroughly in our previous work) [28] and used in [29] for the first time. The anisotropic thermal conductivity is replaced with a parallel and a transversal conductivity with respect to the fiber direction, which is described in the following two equations:

$$K_{\parallel} = K_{\parallel}^f v_f + K_m(1 - v_f) \quad (3)$$

$$K_{\perp} = \frac{K_m(K_m + K_{f\perp} - K_m * v_f + K_{f\perp} * v_f)}{K_m + K_m * v_f + K_{f\perp} - K_{f\perp} * v_f} \quad (4)$$

where $K_{\parallel} \left(\frac{\text{W}}{\text{m} \cdot \text{K}} \right)$ is the parallel thermal conductivity, $K_{\parallel}^f \left(\frac{\text{W}}{\text{m} \cdot \text{K}} \right)$ is the corresponding conductivities of the fibers, $K_{\perp} \left(\frac{\text{W}}{\text{m} \cdot \text{K}} \right)$ is the transversal thermal conductivity, $K_m \left(\frac{\text{W}}{\text{m} \cdot \text{K}} \right)$ is the resin matrix and v_f is the fiber volume fraction and is based on the mixing law theory.

A simplified CFRP structure is thus modeled based on the previous simplification as shown in Fig. 2. The sample is designed to have 16 embedded defects with different shapes and depths. This study considered embedded defects as they better resemble the actual defects in real-life cases as compared to back-drilled defects [30]. In this sample, the size of the defect along the rows increases as you move from left to right, while the depth of the defect increases along the columns as you move from top to bottom keeping the similarly shaped defects at the

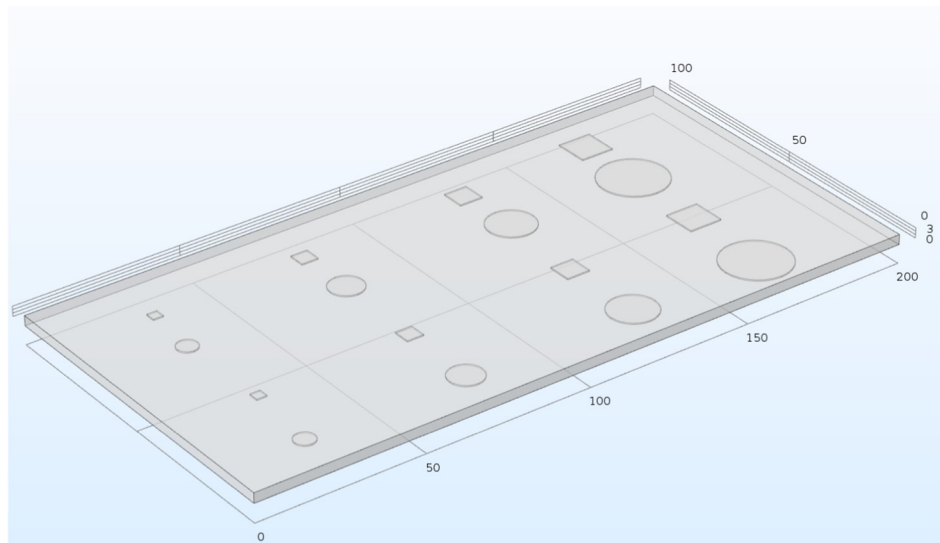


Fig. 2. Simplified CFRP structure with 16 defects.

Table 1
Defects characteristics.

Row #	Depth	Shape	Defect size
1	0.25 mm	Square	Side = 3 mm, 5 mm, 7 mm, 10 mm
2	0.50 mm	Circular	Diameter = 3 mm, 5 mm, 7 mm, 10 mm
3	1.00 mm	Square	Side = 3 mm, 5 mm, 7 mm, 10 mm
4	2.00 mm	Circular	Diameter = 3 mm, 5 mm, 7 mm, 10 mm

same size. Table 1 summarizes the defects characteristics including the shape, size, and depth.

Presented model takes into account all the three heat transfer mechanisms: radiation, conduction, and convection as illustrated in Fig. 3. Radiation occurs between the heat source and the CFRP sample, conduction across the CFRP sample and lastly convection between the CFRP sample and the ambient. Table 2 displays the thermal and physical properties of the CFRP cited from literature [29,31].

A thermogram from the simulated model at $t = 10$ s is shown in Fig. 4. It can be observed that the third and fourth rows are less visible

as compared with the first two; this is because the defects are deeper in these rows. However, considering the thermal contrast instead of temperature will prove more helpful in extracting the depth information as will be shown later in the text.

Temperature curves can then be extracted from the defective areas and subtracted from the sound area curves, in order to obtain the thermal contrast as described in Eq. (5). Thermal contrast is a better feature to consider as it highlights the difference in temperature between the defective and sound areas, for each of the different defects across the sample. Fig. 5 shows a possible location for the sound and defective areas.

$T_c(t)$: AbsoluteThermalContrast

$T_d(t)$: Temperatureondefect

$T_s(t)$: Temperatureonsoundsurface

$$T_c(t) = T_d(t) - T_s(t) \quad (5)$$

Temperature curves from the deeper defect selected in Fig. 5 is

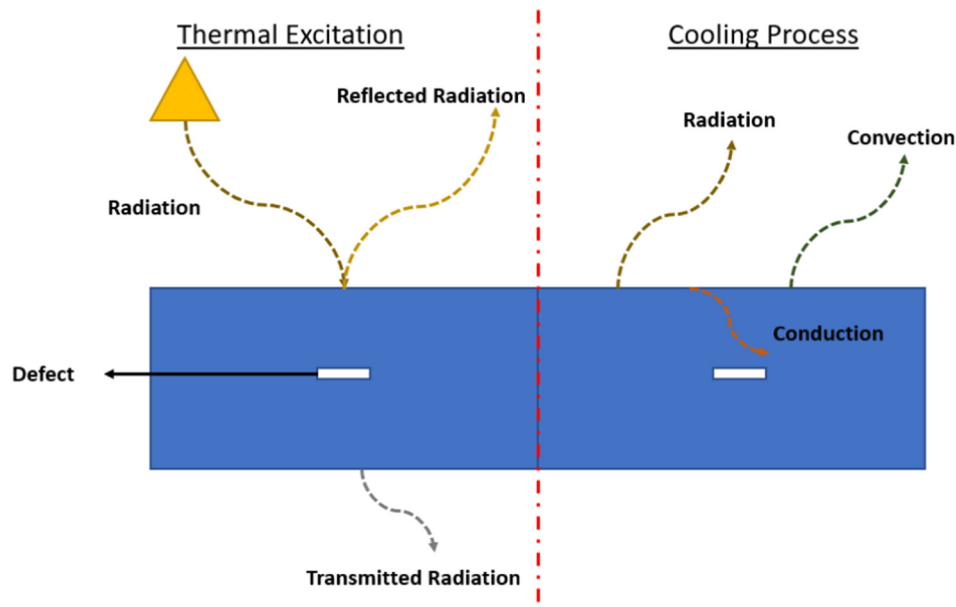


Fig. 3. Heat transfer processes considered during thermal excitation and cooling in the FEM simulations.

Table 2
Thermal and physical parameters values in the conducted simulations.

Symbol	Simulation parameters	Value
ϵ	Emissivity	0.98
ρ_{CFRP}	Density (CFRP)	1500 $\frac{kg}{m^3}$
$C_{p,CFRP}$	Specific heat (CFRP)	1000 J/(kgK)
K_{CFRP}	Thermal Conductivity (fiber)	7 W/(mK)
K_{CFRP}	Thermal Conductivity (fiber $\perp$)	0.8 W/(mK)
ρ_T	Density (air)	1.225 kg/m ³
$C_{p,T}$	Specific heat (air)	1005 J/(kgK)
K_T	Thermal Conductivity (air)	0.026 W/(mK)
h	Convection heat transfer	9.1
L	Length of the Specimen	200 mm
W	Width of the Specimen	100 mm
H	Height of the Specimen	3 mm
T_{amb}	Ambient Temperature	293.15 K
T_o	Initial Temperature	293.15 K
h	Distance between the heat rod and Specimen	7 cm
P_{source}	Wattage of the heat source	500 Watts
t_{pulse}	Duration of the heat pulse	10 s

displayed in Fig. 6. Temperature curves for both areas, the defect-free and the defective are plotted. Also, the thermal contrast using Eq. (5) is calculated and then plotted in the same figure. It can be observed that the thermal contrast is a better approach to capture the deviation of temperature between the defect and the sound areas. In other words, using the temperature curves as distinguishable features for different areas on the sample (defective and non-defective) will not serve us well. Thermal contrast is a better feature to be considered and this can be observed clearly from Fig. 6. Additionally, the time range is an important parameter to consider. The time range in which the thermal contrast curves for different defects are distinguishable is useful; beyond that, if the thermal contrast curves are not distinguishable anymore (due to thermal equilibrium), curves should be truncated before feeding them into the neural network. Note that the temperature curve from Fig. 5 shows the center of the defect for illustration purposes only. However, the actual data collected for training the neural network was obtained by selecting multiple points (5 point locations) per defect, since the temperature across the defective area is not even.

Two interesting cases are highlighted below and depicted in Fig. 7. The first case is examining the temperature curves of the defects in the first row, which have the same shape and depth, but have different sizes. It can be observed from Fig. 8 that the largest defect causes a higher build-up of heat and thus a higher peak temperature is recorded.

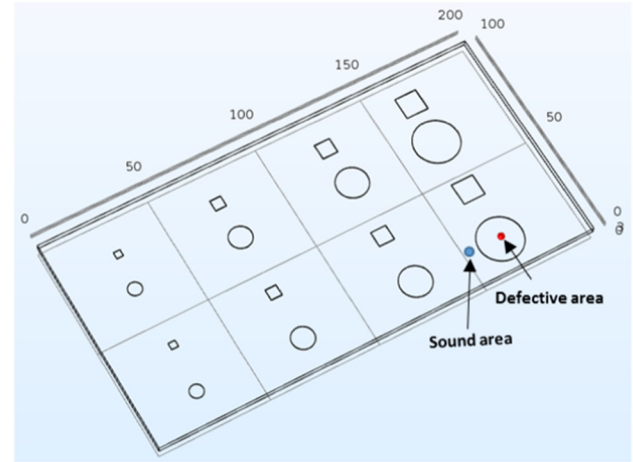


Fig. 5. an example of sound and defective locations.

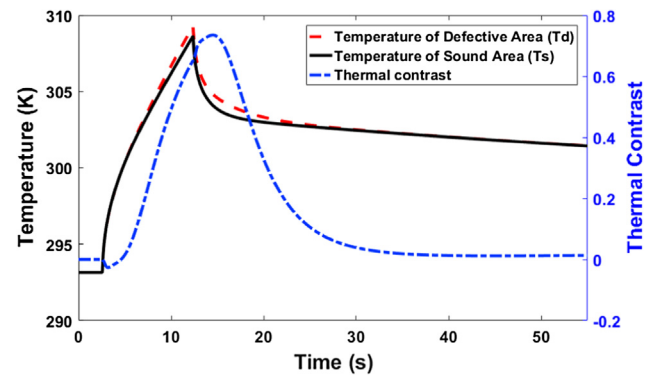


Fig. 6. Temperature and thermal contrast curves.

The second case is examining the temperature curves of defects in a single column, which is column 2. Because, two different shapes are embedded per column, the temperature curves are plotted in the two subplots as shown in Fig. 9. Although the defects of the same shape are of the same size as well, it can be observed that the shallower the defect, the higher its peak temperature becomes. This is due to the thermal resistance near the surface of the shallower defect, which is higher and thus causing a higher heat build-up. This trend in the temperature curves proved useful and will be used in order to detect the defect depth

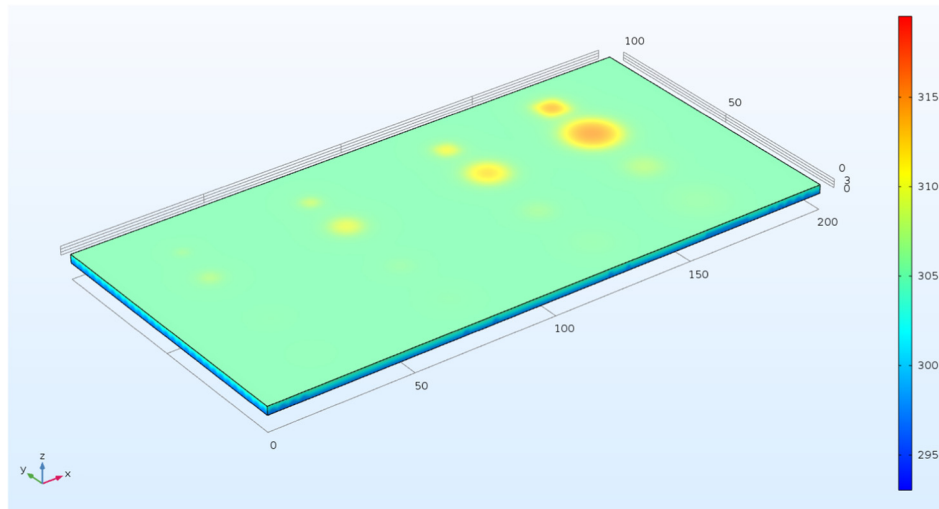


Fig. 4. Simulated thermogram for the CFRP sample at $t = 10$ s.

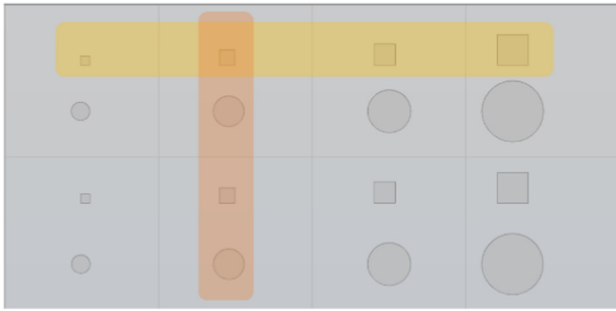


Fig. 7. Defects in row 1 and column 2 which we highlight and compare their thermal curves.

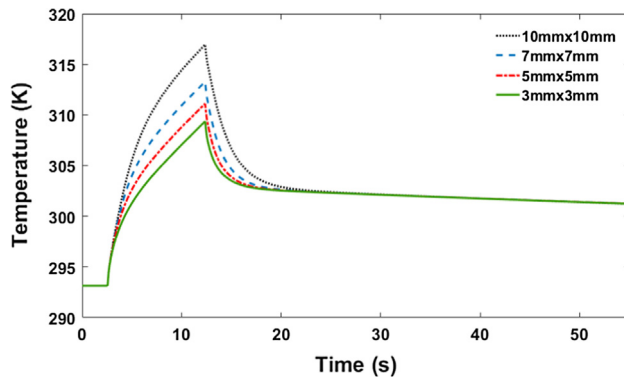


Fig. 8. Temperature curves of the defects in the first row.

using the proposed Neural Network.

4. Neural network

Nowadays, Neural Network based processing is being used as a powerful tool for solving challenging problems in various research fields. Without exceptions, Neural Network was utilized in solving non-linear problems in the thermography field, where analytical models are hard to realize. The working principle of the neural network is inspired by the way our brain works [32]. A neural network consists of nodes and weighed interconnections between these nodes. The input of the network is the thermal contrast curves generated from our FEM simulations.

While selecting points from the simulated thermogram to extract

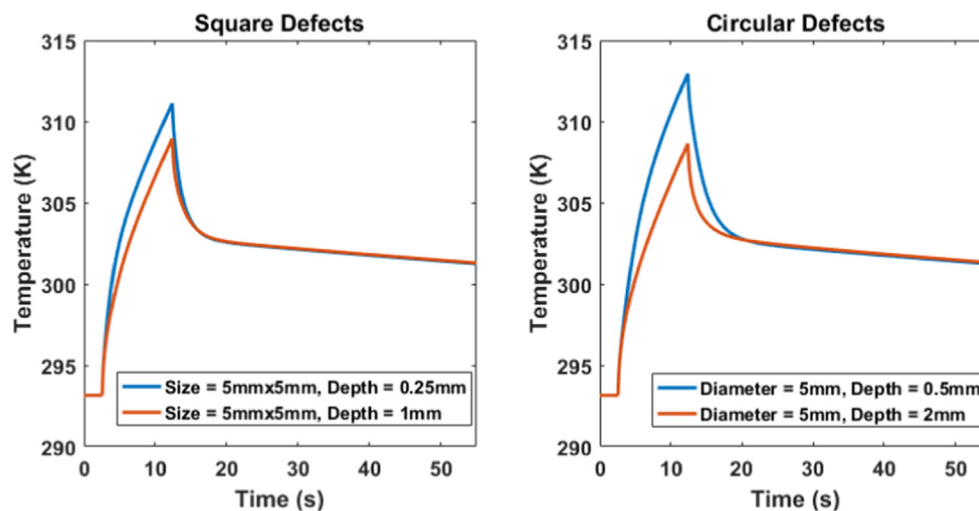


Fig. 9. Temperature curves of the defects in the second column.

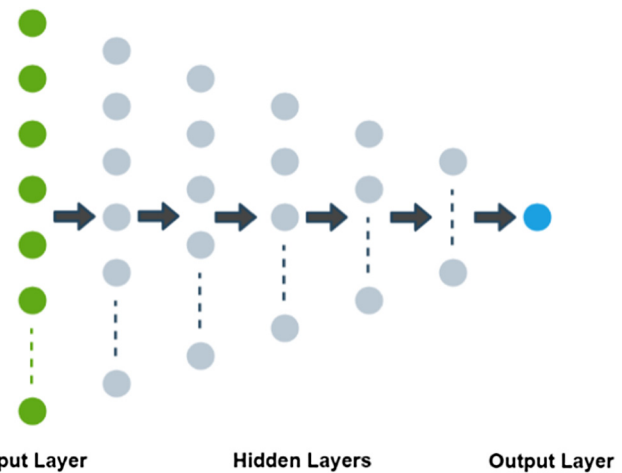


Fig. 10. Architecture of the designed neural network based on the Deep Feed Forward topology.

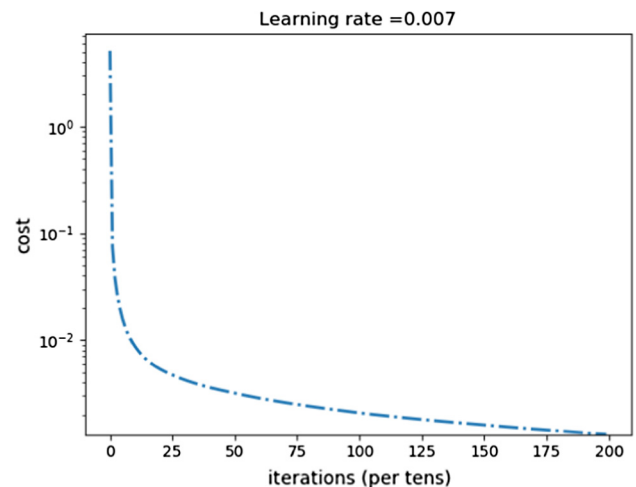


Fig. 11. Decaying cost curve with a learning rate of 0.007.

temperature curves, 5 different locations across the surface above each of the 16 defective areas were considered. This is to account for the small temperature variation across the defect surface. The thermal contrast curves for each of the locations on the surface of defects were

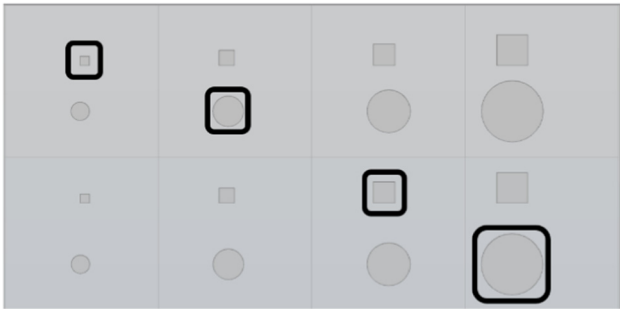


Fig. 12. Selected defects for testing the trained neural network.

Table 3
The estimated depth for the defects located diagonally in the sample.

Test#	Expected output	Estimated output
1	0.250 mm	0.249 mm
2	0.500 mm	0.486 mm
3	1.000 mm	1.004 mm
4	2.000 mm	2.001 mm

calculated using Eq. (5) as explained before. The data is noiseless as it is obtained from the simulation.

In this work, a deep feedforward neural network, also called multilayer perceptron (MLP), is utilized. Feedforward network is used to approximate some function f . In $y = f(x; \theta)$, feedforward neural network will learn the value of the parameters θ (weights) to define mapping between y (output) and x (input) [33]. Our feedforward network consists of input layer, five hidden layers, and an output layer. The input layer has a number of neurons which is equal to the number

of data points of the thermal contrast curves. This depends on the thermal camera acquisition rate/simulation time step. The five hidden layers were designed such that the succeeding layer will have two thirds the number of neurons of that from the preceding layer. The output layer consists of one node which produces the estimated depth. The activation functions of all neurons in the hidden layers are Rectified Linear Unit (ReLU) while a linear activation function was used for the output neuron. The architecture of the network is shown in Fig. 10. The selection of the number of hidden layers is a compromise between accuracy and speed. If you increase the number of hidden layers in the neural network, it will increase the accuracy of detecting deeper defects but will, in turn, increase the training time and become more susceptible to overfitting. Mean square error was chosen as a cost function at the output layer, where the output is the estimated depth as shown in the equation below:

$$cost = \frac{\sum_{i=1}^m (y - \hat{y})^2}{m} \tag{6}$$

where m is the batch size (number of training curves), y is the true depth and $\hat{y}$ is the estimated depth. In addition to the number of hidden layers, other hyper-parameters (learning rate and number of iterations), were set to 0.007 and 5000 respectively. The input training vectors are simply the thermal contrast vectors, which were obtained from the defective areas. As mentioned earlier, for each of the defective locations, where 5 locations across the defect were considered and their thermal contrast curves are obtained. Each thermal contrast curve forms a vector for training. As the training dataset is limited in size, we trained the network with larger number of iterations. A Neural Networks model is said to have overfitted when it performs well on the training set and perform poorly on test set. Several techniques can be used to avoid overfitting, one of this is the cross-validation method which we employed in this study. In cross-validation, we first shuffled

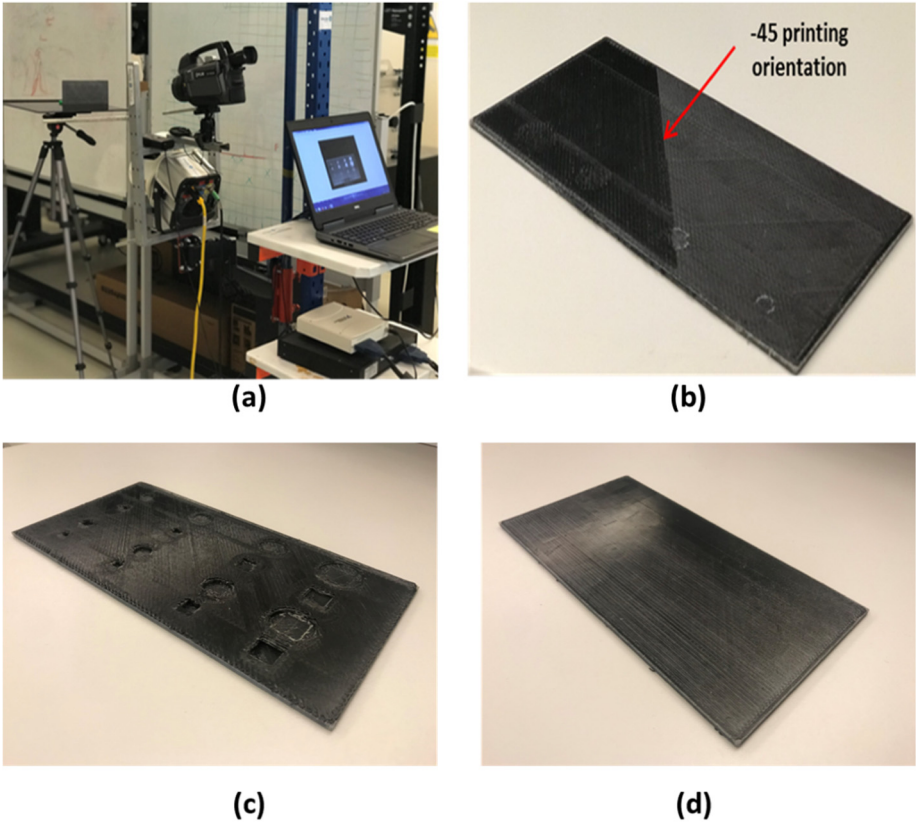


Fig. 13. (a) Pulsed thermography setup (b) 3D printed CFRP sample with embedded defects (c) 3D printed CFRP sample with back-drilled defects (bottom view) (d) 3D printed CFRP sample with back-drilled defects (top view).

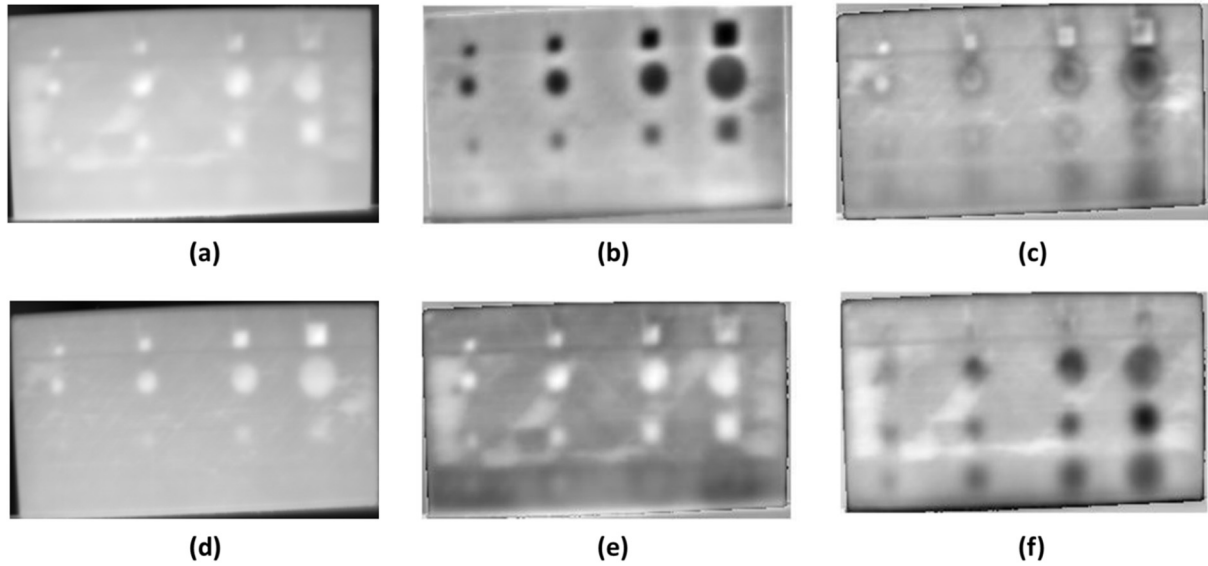


Fig. 14. Obtained thermograms (a) Before enhancement (embedded) (b) PC3 (embedded) (c) PC4 (embedded) (d) Before enhancement (back-drilled) (e) PC3 (back-drilled) (f) PC4 (back-drilled).

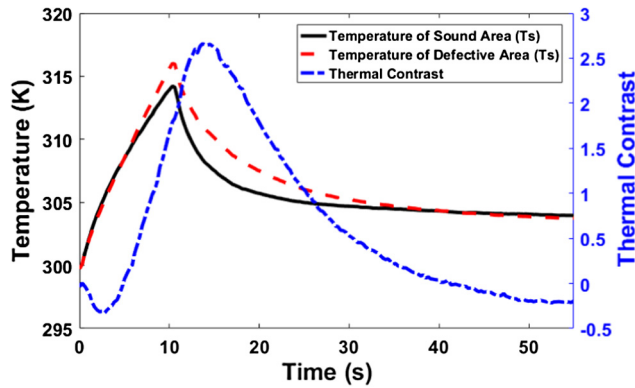


Fig. 15. Temperature curves and thermal contrast from the experimentally acquired thermogram for one of the defects.

and split the simulation data (thermal contrast curves) such that 75% of it was used to train the developed NN while the remaining 25% was used for testing the accuracy of the NN.

The input training data (the thermal contrast curves) was standardized by subtracting the mean and dividing by the standard deviation using Eq. (7), where μ_{T_c} and σ_{T_c} represent the mean and the standard

Table 4

The estimated depth for the defects located diagonally in the sample.

Test#	Expected output	Estimated output (Sample 1)	Estimated output (sample 2)
1	0.250 mm	0.218 mm	0.288 mm
2	0.500 mm	0.514 mm	0.598 mm
3	1.000 mm	0.918 mm	1.018 mm
4	2.000 mm	2.024 mm	2.041 mm

deviation of thermal contrasts, respectively.

$$\widehat{T_{c,n}^i}(t) = \frac{\widehat{T_c}(t) - \mu_{T_c}}{\sigma_{T_c}} \quad (7)$$

Fig. 11 shows the cost curve during the learning process, it can be observed that it is decreasing as the number of iterations increases.

Defects along the diagonal of the sample as depicted in Fig. 12 were selected for testing the neural network, as it contains both the shallowest and deepest defect in the sample. Table 3 shows the estimated depth, and it was observed that the neural network is estimating the defect depth accurately. The calculated accuracy for the worst-case scenario was more than 97% as the data is noiseless (deterministic conditions).

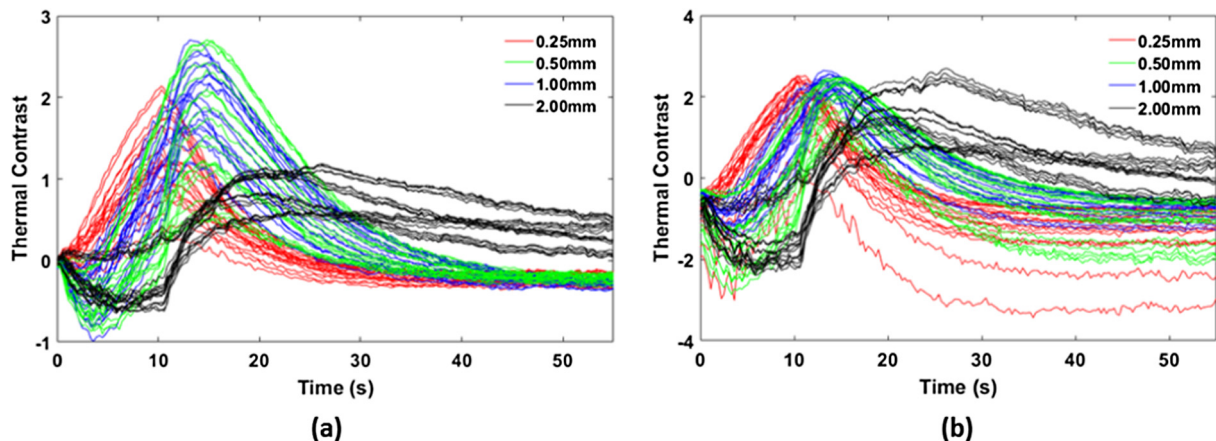


Fig. 16. (a) Thermal contrast curves for all the selected defective locations (b) Standardized thermal contrast curves for all.

5. Experimental validation

The Pulsed Thermography (PT) setup consists of a thermal agitation source, a thermal detector, and mounting/fixating arms. In this work, a halogen lamp rated at 230 V and 2.5 kW was used, along with an Indium Antimonide InSb cooled detector (GF309 from FLIR®), which was set at an acquisition rate of 25 HZ (due to the low conductivity of CFRP), operative in the 3–5 μm range of the IR spectra; with a Noise Equivalent Temperature Difference NETD of 0.025 K and a spatial resolution of $0.4 \times 0.4 \text{ mm/pixel}$. Thermal pulses were controlled and sequenced using an automation box by National Instruments® (NI-MAX). Fig. 13a displays the PT setup fixture as used in this study. Two CFRP samples were designed and 3D printed to experimentally validate the use of NN with thermograms acquired in non-ideal environment (thermal noise). We utilized Mark-Two 3D printer by Markforged® to print both samples which are shown in Fig. 13. The first sample shown in Fig. 13b contains embedded defects (air pockets) and has the same structure, defects number and defects characteristics of the sample designed for simulations, shown in Fig. 2 and tabulated in Table 1. The second CFRP printed sample which is shown in Fig. 13c, d contains flat bottom holes defects (back-drilled) which have different thermal response when compared to the embedded defects. We designed the experimental validation procedure in such a way to have defects of: different type (embedded, back-drilled), different shapes (circular, square), four different sizes and at four different depths. More details about the process of printing 3D composite samples can be found in our previous work [30].

A heat pulse of 10 s duration was deposited on the CFRP sample and a sequence of thermograms for a duration of 55 s was then recorded. The experimental conditions are as follows: the experiments were performed under room temperature of (25 °C) with a normal humidity level of 45%. The distance between the sample and the camera was around 8 cm. Principal Component Analysis PCA was utilized to confirm the existence of the defects. Fig. 14 shows the 3rd and the 4th Principal Components or PC3 and PC4 for both of the samples with the original (raw) thermograms before applying PCA.

Fig. 15 shows the extracted temperature curves for both; a sound and a defective-area for one of the defects. Thermal contrast is then calculated using Eq. (5) and plotted in the same figure. This process is repeated for all of the defects, while selecting 5 different points on the surface above the defects to account for the small variation of temperature across the defective area. Thus, each defect presents 5 different readings.

Fig. 16a displays all the thermal contrast curves, where each bundle of curves corresponds to a different depth. The thermal contrast data is standardized using Eq. (6). The standardization process ensures that the thermal contrast curves are within the same vertical range. The difference in depth is recognized by the amount of drift along the x-axis; deeper defects will manifest heat buildup later in time as compared to the shallower ones as shown in Fig. 16b.

The standardized thermal contrast data is used to train the neural network. As explained in the simulation section, the cross-validation method is used for the testing of the trained neural network. In both samples, the defects along the diagonal as shown in Fig. 12 were used for testing. Table 4 documents the estimated depths by the neural network, where sample 1 refers to the sample containing embedded defects and sample 2 refers to the sample containing back-drilled defects. It was found that the accuracy of this network is nearly 90% considering the worst case. This drop in the network accuracy in estimating the defect depth is attributed to the non-idealities in the experimental setup including the thermal noise in the ambience and humidity. To design a neural network, which is more robust to the experimental non-idealities, larger training dataset should be acquired with different environmental conditions.

6. Conclusion

This work addressed the complex and nonlinear problem of estimating defect depths in anisotropic composites undergoing a thermography testing, using a feedforward NN approach. Pulse-thermography is typically used to suggest a one-dimensional heat transfer in the sample depth, however, for a plastic based composite with high anisotropy; this yields such technique to be equivalent to steady heating, which renders the one-dimensional heat transfer assumption impractical. The proposed methodology of employing an artificial neural network coupled with pulsed thermography is described in details and in two folds; training and setting up the NN (with 5 hidden layers) based on simulated data sets, generated from a COMSOL® multi-physics model. The simulated finite-element model is complete and does describe the heating physics and the composites equivalent thermal properties. The simulated sample provides a deterministic environment for the NN training and setting it up. This first fold provided a worst case scenario 97% accuracy; while the second fold used an experimental PT scheme provided a worst case scenario accuracy of 90%. For future work, the experimental noise may be reduced and filtered; additionally, a zoned approach to training the NN can be used to improve the NN accuracy, where each depth layer (a bundle of laminates is trained separately). Moreover, other NN types and training approaches might be evaluated for their sensitivities, ease of training and accuracy outcome.

Conflict of interest

The authors declared that there is no conflict of interest.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.infrared.2019.03.014>.

References

- [1] X.P. Maldague, Introduction to NDT by active infrared thermography, *Mater. Eval.* 60 (2002) 1060–1073.
- [2] N.P. Avdelidis, B.C. Hawtin, D.P. Almond, Transient thermography in the assessment of defects of aircraft composites, *NDT E Int.* 36 (2003) 433–439.
- [3] D. Hull, T.W. Clyne, *An Introduction to Composite Materials*, Cambridge University Press, 1996.
- [4] S. Sfarra, C. Ibarra-Castaneda, N.P. Avdelidis, M. Genest, L. Bouchagier, D. Kourousis, et al., A comparative investigation for the nondestructive testing of honeycomb structures by holographic interferometry and infrared thermography, *J. Phys. Conf. Ser.* 214 (2010) 012071.
- [5] A. Dillenz, T. Zweschper, G. Busse, Progress in ultrasound phase thermography. *Thermosense XXIII, Int. Soc. Opt. Photon.* 4360 (2001) 574–580.
- [6] M. Grosso, J.E. Lopez, V.M. Silva, S.D. Soares, J.M. Rebello, G.R. Pereira, Pulsed thermography inspection of adhesive composite joints: computational simulation model and experimental validation, *Compos. Part B Eng.* 106 (2016) 1–9.
- [7] D.M. McCann, M.C. Forde, Review of NDT methods in the assessment of concrete and masonry structures, *NDT E Int.* 34 (2001) 71–84.
- [8] T. Li, D.P. Almond, D.A.S. Rees, Crack imaging by scanning pulsed laser spot thermography, *NDT E Int.* 44 (2011) 216–225.
- [9] C. Bonavolonta, M. Valentino, G. Peluso, A. Barone, Non destructive evaluation of advanced composite materials for aerospace application using HTS SQUIDS, *IEEE Trans. Appl. Supercond.* 17 (2007) 772–775.
- [10] A. Manohar, F.L. di Scalea, Modeling 3D heat flow interaction with defects in composite materials for infrared thermography, *NDT E Int.* 66 (2014) 1–7.
- [11] N. Rajic, Principal component thermography for flaw contrast enhancement and flaw depth characterisation in composite structures, *Compos. Struct.* 58 (2002) 521–528.
- [12] J.-F. Vandenrijt, N. Lièvre, M.P. Georges, Improvement of defect detection in shearography by using Principal Component Analysis, *Interferom. XVII Tech. Anal.*, vol. 9203, International Society for Optics and Photonics, 2014, p. 92030L.
- [13] S. Marinetti, E. Grinzato, P.G. Bison, E. Bozzi, M. Chimenti, G. Pieri, et al., Statistical analysis of IR thermographic sequences by PCA, *Infrared Phys. Technol.* 46 (2004) 85–91.
- [14] L. Cheng, B. Gao, G.Y. Tian, W.L. Woo, G. Berthiau, Impact damage detection and identification using eddy current pulsed thermography through integration of PCA and ICA, *IEEE Sens. J.* 14 (2014) 1655–1663.
- [15] M. Omar, M.I. Hassan, K. Saito, R. Alloo, IR self-referencing thermography for

- detection of in-depth defects, *Infrared Phys. Technol.* 46 (2005) 283–289.
- [16] S.M. Shepard, J.R. Lhota, B.A. Rubadeux, D. Wang, T. Ahmed, Reconstruction and enhancement of active thermographic image sequences, *Opt. Eng.* 42 (2003) 1337–1343.
- [17] S.M. Shepard, *Advances in pulsed thermography*, Thermosense XXIII International Society for Optics and Photonics, 2001, pp. 511–516.
- [18] Q. Feng, B. Gao, P. Lu, W.L. Woo, Y. Yang, Y. Fan, et al., Automatic seeded region growing for thermography debonding detection of CFRP, *NDT E Int.* 99 (2018) 36–49.
- [19] S.K. Lau, D.P. Almond, P.M. Patel, Transient thermal wave techniques for the evaluation of surface coatings, *J. Phys. Appl. Phys.* 24 (1991) 428.
- [20] J.-C. Krapez, X. Maldague, P. Cielo, Thermographic nondestructive evaluation: data inversion procedures, *Res. Nondestruct. Eval.* 3 (1991) 101–124.
- [21] J. Peeters, G. Arroud, B. Ribbens, J.J.J. Dirckx, G. Steenackers, Updating a finite element model to the real experimental setup by thermographic measurements and adaptive regression optimization, *Mech. Syst. Signal Process.* 64 (2015) 428–440.
- [22] J. Peeters, G. Steenackers, B. Ribbens, G. Arroud, J. Dirckx, Finite element optimization by pulsed thermography with adaptive response surfaces, *QIRT 2014 12th Int. Conf. Quant. Infrared Thermogr.* 7–11 July Bordx. (2014).
- [23] J.G. Sun, Analysis of pulsed thermography methods for defect depth prediction, *J. Heat Transf.* 128 (2006) 329–338.
- [24] X. Han, L.D. Favro, P.K. Kuo, R.L. Thomas, Early-time pulse-echo thermal wave imaging, *Rev. Prog. Quant. Nondestruct. Eval.* Springer, 1996, pp. 519–524.
- [25] Z. Zeng, J. Zhou, N. Tao, L. Feng, C. Zhang, Absolute peak slope time based thickness measurement using pulsed thermography, *Infrared Phys. Technol.* 55 (2012) 200–204.
- [26] C. Ibarra-Castanedo, X. Maldague, Pulsed phase thermography reviewed, *Quant. Infrared Thermogr. J.* 1 (2004) 47–70.
- [27] F. Lopez, Nicolau V de Paulo, C. Ibarra-Castanedo, X. Maldague, Thermal-numerical model and computational simulation of pulsed thermography inspection of carbon fiber-reinforced composites, *Int. J. Therm. Sci.* 86 (2014) 325–340.
- [28] N. Saeed, M.A. Omar, Y. Abdulrahman, A neural network approach for quantifying defects depth, for nondestructive testing thermograms, *Infrared Phys. Technol.* (2018).
- [29] M.W. Pilling, B. Yates, M.A. Black, P. Tattersall, The thermal conductivity of carbon fibre-reinforced composites, *J. Mater. Sci.* 14 (1979) 1326–1338.
- [30] N. Saeed, M.A. Omar, Y. Abdulrahman, S. Salem, A. Mayyas, IR thermographic Analysis of 3D printed CFRP reference samples with back-drilled and embedded defects, *J. Nondestruct. Eval.* 37 (2018) 59.
- [31] F. Khodayar, F. Lopez, C. Ibarra-Castanedo, X. Maldague, Optimization of the inspection of large composite materials using robotized line scan thermography, *J. Nondestruct. Eval.* 36 (2017) 32.
- [32] Book FAB, *The Handbook of Brain Theory and Neural Networks*, 2002.
- [33] I. Goodfellow, Y. Bengio, A. Courville, Y. Bengio, *Deep Learning* vol. 1, MIT press Cambridge, 2016.